Bhuvan Veeravalli

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Education

University of Alberta Edmonton, AB
BSc Computing Science Specialization, GPA: 3.8/4.0 *Expected May 2027*

- **Relevant Courses:** Machine Learning I & II, Reinforcement Learning, NLP, Algorithms I, Database Management, Software Engineering, Applied Statistics I & II, Linear Algebra I & II

Experience

FES Summer Intern Apr 2026 – Present
Future Energy System *Edmonton, AB*

- Developed spatiotemporal deep learning models to predict 3D thermal dynamics in molten salt energy storage systems.
- Processed infrared imaging data to extract thermal patterns, temperature distributions, and heat-transfer behavior.
- Built ML workflows for thermal forecasting, model validation, and performance evaluation on experimental data.

ML Research Assistant Jan 2025 – Present
Renewable Thermal Energy Lab *Edmonton, AB*

- Developed multi-head attention LSTM model to predict 10-point thermal storage temperature profiles, achieving $R^2 = 0.98$ and MAE $< 0.5°$C across 150+ experimental configurations.
- Engineered automated data ingestion pipeline integrating real-time sensor streams, reducing model validation time by 40%.
- Implemented adaptive error-tracking system to identify optimal test configurations and performance thresholds.

Undergraduate Teaching Assistant, STAT 252 Sep 2025 – Apr 2026
University of Alberta *Edmonton, AB*

- Evaluated 250+ student submissions on statistical modeling and data analytics, applying consistent grading rubrics.
- Provided one-on-one academic support, clarifying regression techniques and data-driven decision-making methodologies.

Projects

LAIrner | *Socratic AI-Powered Tutoring Platform – Next.js, Prisma, Turso, GPT-4o* May 2026 – June 2026

- Identified that conventional AI tutors act as direct answer engines that eliminate cognitive friction, and designed a Socratic-method alternative in response.
- Architected a dual-model LLM orchestration pipeline pairing GPT-4o for multi-turn Socratic reasoning with a zero-shot GPT-4o-mini classifier that routes queries across 14 academic domains in real time.
- Engineered prompt guardrails to block direct answer leakage and built a stateful Next.js/Prisma/Turso backend with an adaptive feedback loop that re-tunes hint density per student.

IntelliDocs | *Document Intelligence System – Mistral-7B, RAG, QLoRA, Hybrid Retrieval*

- Tackled the problem of organizations struggling to surface knowledge buried across thousands of unstructured documents, causing information silos and slow decisions.
- Architected a RAG system on Mistral-7B combining dense (BGE) and sparse (BM25) retrieval, then fine-tuned it with QLoRA on synthetic and human-annotated Q&A pairs.
- Achieved 0.89 NDCG@10 on retrieval quality, improved answer relevance by 35%, and cut hallucination rate by 40%.

Technical Skills

Languages: Java, JavaScript, Python, C/C++, C#, SQL
Frameworks: React, Node.js, Express.js, Spring Boot, FastAPI, Django, Flask, .NET
Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Vector Databases
Cloud & DevOps: AWS (Lambda, S3, RDS, Cognito, Bedrock), Docker, Kubernetes, GCP, CI/CD, Git
AI/ML & GenAI: PyTorch, TensorFlow, LLMs, RAG, LangChain, NLP, Embeddings, Prompt Engineering

Certifications & Awards

Certifications: Generative AI for Software Development (DeepLearning.AI, Oct 2025); Google Cloud Data Analytics Certificate (Google, Sep 2025); Introduction to Data Analytics (IBM, Jul 2025)
Awards: URI Stipend Award – Future Energy Systems (2026); NSERC Undergraduate Student Research Award (2025); Dean's Honor Roll (2024, 2025); University of Alberta Gold Standard & International Admission Scholarships (2023)